



# Synthetic Analytics & Alerting

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**Series:** SYNTH | **Notebook:** 6 of 6 | **Created:** December 2025 | **Last Updated:** 01/28/2026

## Dashboards, SLOs, and Alerting Strategies

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This notebook covers advanced analytics for synthetic monitoring, including building dashboards, configuring SLOs, and implementing effective alerting strategies using the latest Dynatrace platform.

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## Prerequisites

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-  Access to a Dynatrace environment with Synthetic Monitoring
-  Completed SYNTH-01 through SYNTH-05
-  Active synthetic monitors generating data

# 1. Analytics Overview

## Key Metrics for Synthetic Monitoring

| Metric Category | Metrics                     | Use Case        |
|-----------------|-----------------------------|-----------------|
| Availability    | Success rate, failure count | SLA compliance  |
| Performance     | Response time, TTFB, DNS    | User experience |
| Reliability     | Consecutive failures, MTTR  | Stability       |
| Geographic      | Per-location metrics        | Regional issues |

## Data Sources

| Source                       | Description         | Best For            |
|------------------------------|---------------------|---------------------|
| bizevents                    | Execution results   | Real-time analysis  |
| dt.entity.synthetic_test     | Monitor definitions | Configuration       |
| dt.entity.synthetic_location | Location info       | Geographic analysis |
| metrics                      | Aggregated metrics  | Long-term trends    |

# 2. Availability Analysis

## Calculating Availability

Availability = (Successful Executions / Total Executions) × 100%

Example SLA Targets:

- 99.9% = max 8.76 hours downtime/year

- 99.5% = max 43.8 hours downtime/year

- 99.0% = max 87.6 hours downtime/year

```
// Overall synthetic availability (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}
| fieldsAdd availability_pct = round((successful * 100.0) / total_executions,
decimals: 3)
| fieldsAdd downtime_minutes = round((failed * 5.0), decimals: 0) //
Assuming 5-min frequency
```

```
// Availability by monitor (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_test}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
3)
| sort availability_pct asc
| limit 30
```

```
// Availability trend over time
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_count = count()
}, by: {hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) / total_count,
decimals: 2)
| sort hour_bucket desc
```

```
// Daily availability report
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd day = formatTimestamp(timestamp, format: "yyyy-MM-dd")
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {day}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
3)
| sort day desc
| limit 30
```

### 3. Performance Analysis

#### Performance Metrics

| Metric               | Description          | Good    | Warning   | Critical |
|----------------------|----------------------|---------|-----------|----------|
| <b>Response Time</b> | Total execution time | < 2s    | 2-5s      | > 5s     |
| <b>DNS</b>           | DNS resolution       | < 50ms  | 50-200ms  | > 200ms  |
| <b>Connect</b>       | TCP connection       | < 100ms | 100-300ms | > 300ms  |
| <b>TTFB</b>          | Time to first byte   | < 500ms | 500ms-1s  | > 1s     |

```
// Response time percentiles by monitor
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    p50_ms = percentile(toDouble(synthetic.response_time), 50),
    p75_ms = percentile(toDouble(synthetic.response_time), 75),
    p90_ms = percentile(toDouble(synthetic.response_time), 90),
    p95_ms = percentile(toDouble(synthetic.response_time), 95),
    p99_ms = percentile(toDouble(synthetic.response_time), 99),
    executions = count()
}, by: {dt.entity.synthetic_test}
| sort p95_ms desc
| limit 20
```

```
// Performance trend comparison (this week vs last week)
fetch bizevents, from: now() - 14d
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| fieldsAdd week = if(timestamp > now() - 7d, "This Week", else: "Last Week")
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
}, by: {dt.entity.synthetic_test, week}
| sort dt.entity.synthetic_test, week
```

```
// Timing breakdown analysis
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_dns_ms = avg(toDouble(synthetic.dns_time)),
    avg_connect_ms = avg(toDouble(synthetic.connect_time)),
    avg_ssl_ms = avg(toDouble(synthetic.ssl_time)),
    avg_ttfb_ms = avg(toDouble(synthetic.time_to_first_byte)),
    avg_download_ms = avg(toDouble(synthetic.download_time)),
    avg_total_ms = avg(toDouble(synthetic.response_time))
}, by: {dt.entity.synthetic_test}
| fieldsAdd other_ms = avg_total_ms - avg_dns_ms - avg_connect_ms -
avg_ssl_ms - avg_ttfb_ms - avg_download_ms
| sort avg_total_ms desc
| limit 20
```

```
// Performance anomalies (response time > 2x average)
// First calculate average per monitor, then find executions above threshold
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| fieldsAdd response_ms = toDouble(synthetic.response_time)
| summarize {
    avg_response = avg(response_ms),
    max_response = max(response_ms),
    min_response = min(response_ms),
    executions = count()
}, by: {dt.entity.synthetic_test}
| fieldsAdd deviation_factor = round(max_response / avg_response, decimals:
1)
| filter deviation_factor > 2
| sort deviation_factor desc
| limit 50
```

## 4. Location Comparison

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### Geographic Analysis

Compare synthetic results across locations to:

- Identify regional performance issues
- Detect CDN or DNS problems
- Validate global availability
- Optimize user experience by region

```
// Availability by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_location}
| fieldsAdd availability_pct = round((successful * 100.0) / total, decimals:
2)
| sort availability_pct asc
| limit 30
```

```
// Performance by location
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p50_ms = percentile(toDouble(synthetic.response_time), 50),
    p95_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
}, by: {dt.entity.synthetic_location}
| sort avg_response_ms desc
| limit 30
```

```
// Location performance heatmap (monitor x location)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    availability_pct = countIf(synthetic.availability == true) * 100.0 /
count()
}, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| sort dt.entity.synthetic_test, avg_response_ms desc
```

```
// Location-specific failures
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    unique_monitors = countDistinct(dt.entity.synthetic_test),
    error_types = collectDistinct(synthetic.error_message)
}, by: {dt.entity.synthetic_location}
| sort failure_count desc
| limit 20
```

## 5. SLO Configuration

### Service Level Objectives for Synthetic

| SLO Type     | Metric            | Example Target     |
|--------------|-------------------|--------------------|
| Availability | Success rate      | 99.9%              |
| Performance  | Response time P95 | < 3000ms           |
| Error Budget | Allowed failures  | 0.1% of executions |

### Creating Synthetic SLOs

1. **Navigate to:** Settings → Service-level objectives
2. **Create SLO:** Name, description, timeframe
3. **Define metric:** Use synthetic availability or response time
4. **Set target:** e.g., 99.9% availability
5. **Configure alerting:** Error budget alerts

```
// SLO calculation - Availability (30-day window)
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| summarize {
    total_executions = count(),
    successful = countIf(synthetic.availability == true),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_test}
| fieldsAdd sli_availability = round((successful * 100.0) / total_executions,
decimals: 4)
| fieldsAdd slo_target = 99.9
| fieldsAdd error_budget_total = round(total_executions * 0.001, decimals: 0)
// 0.1% budget
| fieldsAdd error_budget_remaining = round(error_budget_total - failed,
decimals: 0)
| fieldsAdd error_budget_pct = round((error_budget_remaining * 100.0) /
error_budget_total, decimals: 1)
| fieldsAdd slo_status = if(sli_availability >= slo_target, "✅ Met",
else: "❌ Breached")
| sort sli_availability asc
| limit 30
```



```
// SLO calculation - Performance (P95 < 3000ms)
fetch bizevents, from: now() - 30d
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    executions = count(),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    within_target = countIf(toDouble(synthetic.response_time) < 3000)
}, by: {dt.entity.synthetic_test}
| fieldsAdd sli_performance = round((within_target * 100.0) / executions,
decimals: 2)
| fieldsAdd slo_target = 95.0 // 95% of requests < 3s
| fieldsAdd slo_status = if(sli_performance >= slo_target, "✅ Met", else:
"❌ Breached")
| sort sli_performance asc
| limit 30
```

```
// Error budget burn rate (last 7 days)
fetch bizevents, from: now() - 7d
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd day = formatTimestamp(timestamp, format: "yyyy-MM-dd")
| summarize {
    total = count(),
    failed = countIf(synthetic.availability == false)
}, by: {dt.entity.synthetic_test, day}
| fieldsAdd daily_error_rate = round((failed * 100.0) / total, decimals: 3)
| fieldsAdd budget_consumed = round(failed * 100.0 / (total * 0.001),
decimals: 1) // vs 0.1% budget
| sort dt.entity.synthetic_test, day desc
```

## 6. Alerting Strategies

### Alert Types

| Alert Type   | Trigger                  | Use Case          |
|--------------|--------------------------|-------------------|
| Availability | Consecutive failures     | Outage detection  |
| Performance  | Response time threshold  | Degradation       |
| SLO          | Error budget consumption | Proactive warning |
| SSL          | Certificate expiration   | Security          |

## Alert Configuration Best Practices

| Setting                     | Recommendation    | Reason                  |
|-----------------------------|-------------------|-------------------------|
| <b>Consecutive failures</b> | 2-3               | Avoid false positives   |
| <b>Location threshold</b>   | 2+ locations      | Confirm not local issue |
| <b>Alert delay</b>          | 5-10 minutes      | Allow for transients    |
| <b>Auto-resolve</b>         | After 2 successes | Clear resolved issues   |

```
// Monitors with recent failures (potential outages)
// Check for monitors that have failed multiple times recently
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failure_count = count(),
    last_failure = max(timestamp),
    first_failure = min(timestamp)
}, by: {dt.entity.synthetic_test, dt.entity.synthetic_location}
| filter failure_count >= 2
| sort failure_count desc
| limit 20
```

```
// Performance degradation alerts (P95 > threshold)
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    avg_response_ms = avg(toDouble(synthetic.response_time)),
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95),
    executions = count()
}, by: {dt.entity.synthetic_test}
| filter p95_response_ms > 5000 // > 5 seconds threshold
| sort p95_response_ms desc
```

```
// Multi-location failure detection (global outage indicator)
fetch bizevents, from: now() - 30m
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| summarize {
    failing_locations = countDistinct(dt.entity.synthetic_location),
    failure_count = count(),
    locations = collectDistinct(dt.entity.synthetic_location)
}, by: {dt.entity.synthetic_test}
| filter failing_locations >= 2
| fieldsAdd severity = if(failing_locations >= 3, "CRITICAL", else:
"WARNING")
| sort failing_locations desc
```

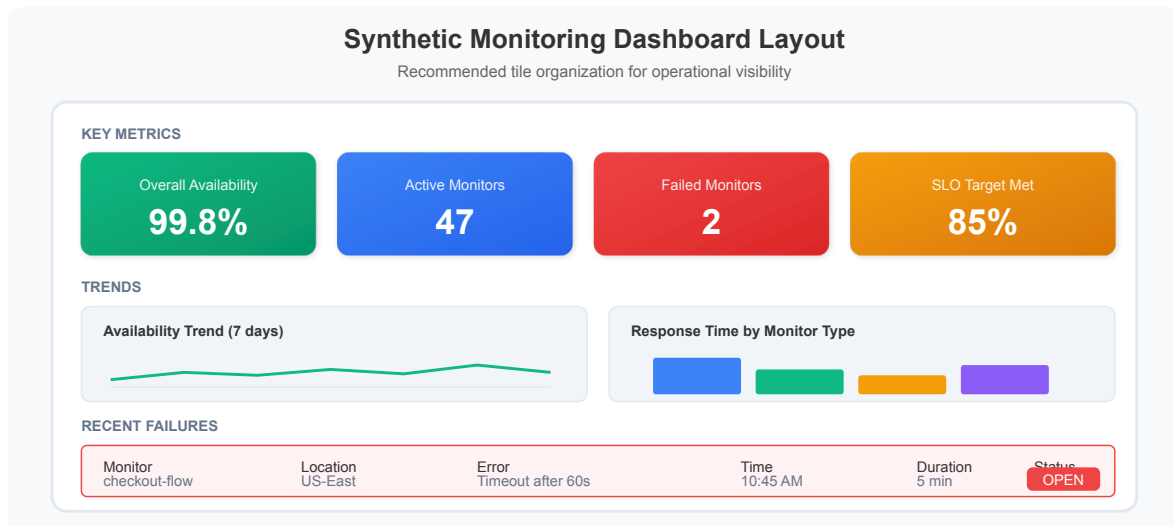
## 7. Dashboard Building

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### Recommended Dashboard Tiles

| Tile Type          | Content                 | Visualization |
|--------------------|-------------------------|---------------|
| Overall Health     | Availability %          | Single value  |
| Availability Trend | Hourly availability     | Line chart    |
| Response Time      | P95 by monitor          | Bar chart     |
| Location Map       | Geographic availability | World map     |
| Failures Table     | Recent failures         | Table         |
| SLO Status         | Error budget            | Gauge         |

## Dashboard Layout



```
// Dashboard tile: Overall availability (single value)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    availability_pct = round(countIf(synthetic.availability == true) * 100.0
/ count(), decimals: 2)
}
```

```
// Dashboard tile: Active vs failed monitors
fetch bizevents, from: now() - 1h
| filter event.provider == "dynatrace.synthetic"
| summarize {
    latest_status = takeFirst(synthetic.availability)
}, by: {dt.entity.synthetic_test}
| summarize {
    total_monitors = count(),
    healthy = countIf(latest_status == true),
    failing = countIf(latest_status == false)
}
```

```
// Dashboard tile: Response time by monitor (bar chart)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == true
| summarize {
    p95_response_ms = percentile(toDouble(synthetic.response_time), 95)
}, by: {dt.entity.synthetic_test}
| sort p95_response_ms desc
| limit 10
```

```
// Dashboard tile: Recent failures table
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| filter synthetic.availability == false
| fields timestamp,
    monitor = dt.entity.synthetic_test,
    location = dt.entity.synthetic_location,
    error = synthetic.error_message
| sort timestamp desc
| limit 20
```

```
// Dashboard tile: Hourly availability trend (line chart)
fetch bizevents, from: now() - 24h
| filter event.provider == "dynatrace.synthetic"
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {
    success_count = countIf(synthetic.availability == true),
    total_executions = count(),
    failures = countIf(synthetic.availability == false)
}, by: {hour_bucket}
| fieldsAdd availability_pct = round((success_count * 100.0) /
total_executions, decimals: 2)
| sort hour_bucket asc
```

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## Summary

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In this notebook, you learned:

- ✅ **Availability analysis** - Calculating and trending availability
- ✅ **Performance analysis** - Percentiles, timing breakdown, anomalies
- ✅ **Location comparison** - Geographic performance analysis

- ✓ **SLO configuration** - Service level objectives and error budgets
  - ✓ **Alerting strategies** - Outage detection, performance alerts
  - ✓ **Dashboard building** - Key tiles and layouts
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## Series Complete!

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You have completed the Synthetic Monitoring Best Practices series:

1. ✓ **SYNTH-01:** Fundamentals
  2. ✓ **SYNTH-02:** Browser Monitors
  3. ✓ **SYNTH-03:** HTTP Monitors
  4. ✓ **SYNTH-04:** Private Locations
  5. ✓ **SYNTH-05:** Network Monitoring
  6. ✓ **SYNTH-06:** Analytics & Alerting
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## References

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- [Synthetic Monitoring Overview](#)
  - [Analyze Synthetic Monitors](#)
  - [Service Level Objectives](#)
  - [Alerting Profiles](#)
  - [Dashboards](#)
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